

Customer, pricing & transportation intelligence

Decision window: 1-30 October 2025. Release 0.4. Built as a synthetic portfolio simulation inspired by electronic toll-road decisions.

The business decision

Choose customer incentives and effective price strategies under spending, contact, inventory, reward and roadway limits. Future contribution on days 31-90 is estimated separately; no incremental twelve-month value is claimed.

Recommended simulation plan

Expected outcome	Result
Additional trips	275.8
Net incremental contribution	CAD 2,007.65
Incentive costs / budget	CAD 1,596.53 / 1,600.00
Customer contacts	180
Constraint validation	All implemented shared constraints passed

All outcome figures above are model-based planning estimates inside a simulation. They are not observed company results. Zones, rates, engineering capacities and business policies are invented. No affiliation with 407 ETR.

What supports the decision

The pipeline contains 8,000 synthetic customers, 1,221,317 trips and 405,632 digital events. Public Toronto weather, Ontario holidays and Bank of Canada exchange rates provide contextual data.

Prediction evidence

Model	Test AUC	Brier score	Test rows
Propensity	0.895	0.124	8,000
Churn	0.889	0.039	6,225
Attrition	0.735	0.149	6,225

Train snapshots end January 2025; validation and calibration use April; the untouched prediction test uses July. Ninety-day label windows are purged. Repeated customers across time match a batch-rescoring use case. Designed synthetic data cannot establish real-world model quality.

Randomized offer experiment

The four groups contain 2,000 customers each. The planned requirement was 1,962 per group for a 5-percentage-point primary response effect, 80% power and family alpha 5%. The planned target was met. Three treatment/control comparisons use Bonferroni adjustment.

Offer	Response difference	Adjusted interval
Off-peak 15%	+12.4 pp	+8.9 to +16.0 pp
Weekend 20%	+15.4 pp	+11.7 to +19.0 pp
500 loyalty points	+9.0 pp	+5.4 to +12.6 pp

A fresh constrained-policy trial randomized 4,000 policy and 4,000 control customers, independently of model training. Estimated incremental net contribution is CAD 0.38 per randomized customer (95% interval -0.39 to 1.16). It includes uncontacted customers; wide uncertainty remains visible.

A constrained, reviewable plan

One Gurobi mixed-integer solve chooses promotions and loyalty awards together. The optimizer can assign at most one offer per customer, enforce eligibility and inventory, reserve full reward liability, meet portfolio ROI and respect cell-level headroom including a 20% forecast reserve.

How targeting methods compare

Method	Contacts	Expected net CAD	Expected trips
Joint optimization	180	2,007.65	275.8
Propensity ranking	180	1,093.60	261.2
Frequency rules	180	1,111.11	262.6

Methods use the same shortlist and constraints. Differences compare planning objectives; adding later contribution can trade immediate contribution against future value.

Limits that affect interpretation

Optimality covers 600 shortlisted customers and three offers. Trip and later-value effects use a declared 25% shrinkage, not a confidence bound. Retention is reported separately to avoid double counting.

Discount costs include baseline travel. A loyalty award reserves CAD 5 plus CAD 0.35 contact cost; balances reconcile earnings plus awards less redemptions. The joint pricing solver shares capacity with campaign contacts and assumes constant elasticity without fitted price/offer interactions.

The serving demand baseline has next-day MAE 10.48; the learned candidate was rejected by its untouched acceptance test. Fixed-origin 30-day backtests have MAE 9.91 and 92.5% marginal interval coverage. Hour/direction forecasts disaggregate daily demand; hourly coverage is unvalidated.

Recommended next decision

Review scenario economics and the fresh policy experiment before operational adoption. Replace invented rules with approved business policies, enforce realized spending/traffic limits and validate performance on the actual deployment population.

Execution evidence and boundaries

Twelve app workspaces include saved scenario audit records, customer reward/consent profiles, price allocation, policy experiments, an operations review queue and visual SQL cases. Production mode requires approved OIDC identities, release integrity and model acceptance.

Delivery status

Deliverable	Actual status
Local analytics and app	Acceptance checks passed; monthly demand backtests and shared constraint checks recorded.
Model registry	Nine model families registered locally; loaded scoring roundtrips matched. Promotion remains reviewed.
Native Power BI	Eight pages, 16 tables and 32 measures structurally validated. Revised Desktop refresh blocked by reload-dialog automation.
AWS / Databricks	Six AWS contracts run locally; rejected candidates held. Full Databricks marts authored. No hosted execution claimed.
Production deployment	Private auth, containers, CI and AWS storage/IAM template prepared. Runtime dependency audit and infrastructure validation passed; hosted acceptance pending.

Source and privacy notes

Public context uses Open-Meteo historical weather, Ontario holiday data from Nager.Date and Bank of Canada FXCADUSD. Customer IDs and all customer behaviour are synthetic; raw public responses and provenance remain local.

Hosted execution, provider sign-in, container builds, staging load/backup tests and GitHub publication need completed external receipts. No AWS resources or charges were created by this release.